Zixuan Guo

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RESEARCH INTERESTS

Music & Audio Representation Learning · Machine Music Understanding · Music Generation · Music Information Retrieval

EDUCATION

New York University — New York, NY
Master of Music (M.M.) in Music Technology — *Sep 2024 – May 2026*
GPA: 3.98/4.0

Jilin University (Project 985 & 211 University) — Jilin, China
Bachelor of Science (B.S.) in Computer Science and Technology — *Sep 2019 – Jun 2024*

Berklee College of Music — Boston, MA
Five-Week Summer Performance Program (Woodwind Performance) — *Jul 2023 – Aug 2023*

PUBLICATIONS & PREPRINTS

- **Guo, Z.**, Fang, Q., Yang, J.-S., & Rhee, M. (2026). *Closed-Loop Audio Neuroregulation for ADHD: Graph-Based Attention Decoding and Rhythm Entrainment*. *IEEE Access*. doi:10.1109/ACCESS.2026.3731552.
- **Guo, Z.**, Zhong, J., & Chen, S. (2025). *Noise-Robust Instrument Classification under Real-World SNR Conditions*. *SSRN Electronic Journal* (Preprint). SSRN:6510838.
- Zhang, F., **Guo, Z.**, Ding, J., Yang, J., & Liu, W. (2026). *Adaptive Sensor Fusion for Robust Perception in Dense Fog: A Gated Vision and LiDAR Integration Framework*. *Sensors*. doi:10.3390/s26123728.
- Mo, T., Zhang, C., Zou, J., **Guo, Z.**, & Rhee, M. (2026). *Self-Evolving AI Agents with Dual Memory for Automated Software Testing and Bug Localization*. *IEEE Access*. doi:10.1109/ACCESS.2026.3713401.

RESEARCH EXPERIENCE

EEG-Based Attention Decoding and Simulated Closed-Loop Audio Neuroregulation — *2026*
First-author research, IEEE Access — Paper

- Developed AL-STGCN, an adaptive spatio-temporal graph neural network integrating EEG electrode topology, spectral features, and temporal attention for neural-state decoding.
- Evaluated the decoder on public EEG datasets, achieving 89.2% subject-independent ADHD/control classification accuracy and 77.1% accuracy on externally validated, behaviorally defined attention lapses.
- Designed ARIA, a simulated closed-loop framework combining reinforcement-learning-based control with adaptive audio modulation; evaluated robustness through leakage-controlled protocols, ablations, and sensitivity analyses across alternative neural-response models.

Master's Thesis — Controllable Symbolic Music Generation for Human-AI Music Creation — *2025–2026*
New York University — Code

- Studied symbolic music representations and musical structure to enable fine-grained, iterative control in DAW-based creative workflows.
- Developed a hybrid framework combining MuseCoco generation, deterministic symbolic constraints, and AR-VAE latent-space editing across New, Continue, and Transformation modes.
- Built a JUCE DAW plugin with an inference backend for non-destructive generation and refinement.
- Evaluated the system through ablation studies, constraint-satisfaction metrics, and a 7-participant user study; improved New-mode constraint success from 48% to 76% (+28 pp).

Noise-Robust Musical Instrument Recognition — *2025*
New York University — Code

- Investigated representation robustness under acoustic distribution shift, evaluating LAION-CLAP embeddings across progressively degraded SNR conditions.

- Constructed controlled noisy variants of Medley-solos-DB using MUSAN noise and developed noise-aware curriculum training for TCN and lightweight Transformer classifiers.
- Achieved 95.93% clean accuracy and 81.28% accuracy at -5 dB SNR; introduced pure-noise negatives and a confidence-triggered harmonic fallback for explicit rejection under extreme noise.

Bachelor's Thesis — Domain-Adversarial Transfer Learning *2024*
Jilin University Code

- Studied unsupervised domain adaptation by introducing class-matched Mixup/CutMixup intermediate domains within a DANN framework.
- Developed a staged source $\rightarrow$ intermediate $\rightarrow$ target adaptation pipeline and evaluated it across Office-31, ImageCLEF, and OfficeHome, improving ImageCLEF accuracy from 76.84% to 95.00%.

ACADEMIC SERVICE

- Peer Reviewer, *Scientific Reports* (Nature Portfolio), 2026 — 2 manuscripts; 4 completed review rounds
- Peer Reviewer, *Discover Artificial Intelligence* (Springer Nature), 2026 — 1 manuscript; 1 completed review

HONORS & AWARDS

- Academic Achievement Recognition, Graduate Music Technology Program, NYU Steinhardt *May 2026*
- Third-Class Scholarship, Department of Computer Science and Technology, Jilin University *Sep 2020*

PROFESSIONAL EXPERIENCE

Research Fellow — Relational AI Evaluation *Sep 2026 – Present*
Trivian Institute Remote

- Investigating reproducibility, falsification, and adversarial evaluation of relational AI systems through claim-driven experiments and failure analysis.
- Developing reproducible test suites and documenting negative or inconclusive findings to support evidence-based evaluation.

Machine Learning Engineer *Jul 2026 – Aug 2026*
APEXUS-TECH Remote

- Built a quantitative-research RAG assistant using LangChain, MiniLM, Chroma, and Llama 3.1, achieving Precision@5 = 0.92 and NDCG@5 = 0.94.
- Developed hybrid retrieval and structured backtesting workflows; identified target leakage in an earnings-prediction pipeline and implemented feature-timing safeguards.

Research Collaborator — DeepSqueak *May 2025 – Nov 2025; Sep 2026 – Present*
University of Washington Code

- Modernized spectrogram-based acoustic event detection by migrating from YOLOv2 to YOLOX and developing noise-robust preprocessing pipelines.
- Designed the DeepSqueakPy migration architecture and reproducible data organization to support future Python-based development.

Licenses & Certifications

- CITI Program — Social & Behavioral Research (*Apr 2025; valid through Apr 2028*)

TECHNICAL SKILLS

Audio & Music AI: Music Information Retrieval, Machine Listening, Audio Representation Learning, Symbolic Music Generation, Audio DSP, Acoustic Event Detection
Machine Learning: PyTorch, Graph Neural Networks (GNNs), Transformers, VAEs, Representation Learning, Domain Adaptation, Reinforcement Learning
Neural Signal Processing: EEG Signal Processing, Spatio-Temporal Modeling, Attention-State Decoding, Time-Frequency Analysis
Audio Software & Creative Computing: JUCE, Max/MSP, SuperCollider, Ableton Live, Pro Tools
Programming: Python, C/C++, MATLAB, SQL
Systems & Tools: Linux, Git, Docker, REST APIs